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Television upgrade

Abstract

Disclosed herein are system, apparatus, article of manufacture, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for upgrading one or more display device hardware components. In one embodiment, as new television hardware components become available, the technology described herein allows replacement of original hardware components by external hardware components implemented as an add-in card, stick or equivalent circuitry and operational instructions. In some embodiments, the technology described herein may implement an application programming interface (API) that exposes hardware of a new external device communicatively coupled to the display device (e.g., television) to take control of one or more components of an existing internal system. Various interface technologies, such as HDMI and USB, may be used to connect the external hardware to the internal systems.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/740,907, filed on May 10, 2022, which is incorporated herein by reference in its entirety.

BACKGROUND

Field

(1) This disclosure is generally directed to display screen technology, and more particularly to upgrading one or more television components.

Background

(2) After a media device (e.g., a television) is purchased by a customer, methods for improving the experience or performance of the television are generally limited to software only updates. For example, as television and communication technology evolves, existing hardware may be unable to connect new physical protocols, decode certain streams that require more processing power or add new features that are not part of the original design or build.

SUMMARY

(3) Provided herein are system, apparatus, article of manufacture, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for upgrading existing display device hardware components. The technology as described herein, in some embodiments, may be configured to improve television performance (e.g., usability, speed, streaming speed, image quality, sound quality, etc.). In some embodiments, the technology may be applied broadly to any configurable aspect of streaming video and audio.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The accompanying drawings are incorporated herein and form a part of the specification.
- (2) FIG. 1 illustrates a block diagram of a multimedia environment, according to some embodiments.
- (3) FIG. 2 illustrates a block diagram of a streaming media device, according to some embodiments.
- (4) FIG. 3A illustrates a block diagram of a display device without a hardware upgrade device connected, according to some embodiments.
- (5) FIG. 3B illustrates a block diagram of a display device with a hardware upgrade device connected, according to some embodiments.
- (6) FIG. 4 illustrates a block diagram of an example display device internal configuration, according to some embodiments.
- (7) FIG. 5A illustrates a block diagram of a display device with a hardware upgrade device not yet operative with the display device, according to some embodiments.
- (8) FIG. 5B illustrates a block diagram of a display device with a hardware upgrade device replacing existing hardware components, according to some embodiments.
- (9) FIG. 5C illustrates a block diagram of a display device with a hardware upgrade device operating with existing hardware, according to some embodiments.
- (10) FIG. 5D illustrates a block diagram of a display device with a hardware upgrade device adding new functionality, according to some embodiments.
- (11) FIG. 6 illustrates a process diagram of adding one or more hardware components, according to some embodiments.
- (12) FIG. 7 illustrates an example computer system useful for implementing various embodiments.
- (13) In the drawings, like reference numbers generally indicate identical or similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

- (14) Provided herein are system, apparatus, article of manufacture, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for upgrading one or more display device hardware components. For example, as new television hardware components become available, the technology described herein allows replacement of original hardware components by external hardware components implemented as an add-in card, stick or equivalent circuitry and operational instructions.
- (15) In some embodiments, the technology described herein may implement an application programming interface (API) that exposes hardware of a new external device communicatively coupled to the display device (e.g., television) to take control of one or more components of an existing internal system.
- (16) In various embodiments, the technology described herein will replace one or more existing internal hardware components of a display device with new corresponding external hardware components, add one or more external hardware components that will work in conjunction with existing internal hardware components or add new functionality to the display device with one or more new external hardware components.
- (17) In some embodiments, the technology described herein may also be configured with a new or upgraded Operating System (OS) in conjunction with the new external hardware components.
- (18) In some embodiments, the technology described herein may be implemented by forwarding or rerouting of existing Hardware Abstraction Layer (HAL) messages and/or data from existing peripherals to a new external hardware device.
- (19) In some embodiments, the technology described herein may implement Peripheral Abstraction Layer (PAL) APIs. For example, one or more components of the original hardware APIs may be

implemented as peripherals (using PAL APIs) based on an underlying Hardware Abstraction Layer (HAL).

(20) Various embodiments of this disclosure may be implemented using and/or may be part of a multimedia environment **102** shown in FIG. **1**. It is noted, however, that multimedia environment **102** is provided solely for illustrative purposes, and is not limiting. Embodiments of this disclosure may be implemented using and/or may be part of environments different from and/or in addition to the multimedia environment **102**, as will be appreciated by persons skilled in the relevant art(s) based on the teachings contained herein. An example of the multimedia environment **102** shall now be described.

(21) Multimedia Environment

(22) FIG. **1** illustrates a block diagram of a multimedia environment **102**, according to some embodiments. In a non-limiting example, multimedia environment **102** may be directed to streaming media. However, this disclosure is applicable to any type of media (instead of or in addition to streaming media), as well as any mechanism, means, protocol, method and/or process for distributing media.

(23) The multimedia environment **102** may include one or more media systems **104**. A media system **104** could represent a family room, a kitchen, a backyard, a home theater, a school classroom, a library, a car, a boat, a bus, a plane, a movie theater, a stadium, an auditorium, a park, a bar, a restaurant, or any other location or space where it is desired to receive and play streaming content. User(s) **132** may operate with the media system **104** to select and consume content.

(24) Each media system **104** may include one or more media devices **106** each coupled to one or more display devices **108**. It is noted that terms such as “coupled,” “connected to,” “attached,” “linked,” “combined” and similar terms may refer to physical, electrical, magnetic, logical, etc., connections, unless otherwise specified herein.

(25) Media device **106** may be a streaming media device, DVD or BLU-RAY device, audio/video playback device, cable box, and/or digital video recording device, to name just a few examples. Display device **108** may be a monitor, television (TV), computer, touch screen, smart phone, tablet, wearable (such as a watch or glasses), virtual reality (VR) headset, appliance, internet of things (IOT) device, automotive display, gaming display, heads-up display (HUD), and/or projector, to name just a few examples. In some embodiments, media device **106** can be a part of, integrated with, operatively coupled to, and/or connected to its respective display device **108**.

(26) Each media device **106** may be configured to communicate with network **118** via a communication device **114**. The communication device **114** may include, for example, a cable modem or satellite TV transceiver. The media device **106** may communicate with the communication device **114** over a link **116**, wherein the link **116** may include wireless (such as WiFi) and/or wired connections.

(27) In various embodiments, the network **118** can include, without limitation, wired and/or wireless intranet, extranet, Internet, cellular, Bluetooth, infrared, and/or any other short range, long range, local, regional, global communications mechanism, means, approach, protocol and/or network, as well as any combination(s) thereof.

(28) Media system **104** may include a remote control **110**. The remote control **110** can be any component, part, apparatus and/or method for controlling the media device **106** and/or display device **108**, such as a remote control, a tablet, laptop computer, smartphone, wearable, on-screen controls, integrated control buttons, audio controls, or any combination thereof, to name just a few examples. In an embodiment, the remote control **110** wirelessly communicates with the media device **106** and/or display device **108** using cellular, Bluetooth, infrared, etc., or any combination thereof. The remote control **110** may include a microphone **112**, which is further described below.

(29) The multimedia environment **102** may include a plurality of content servers **120** (also called content providers or sources **120**). Although only one content server **120** is shown in FIG. **1**, in practice the multimedia environment **102** may include any number of content servers **120**. Each

content server **120** may be configured to communicate with network **118**.

(30) Each content server **120** may store content **122** and metadata **124**. Content **122** may include any combination of music, videos, movies, TV programs, multimedia, images, still pictures, text, graphics, gaming applications, advertisements, programming content, public service content, government content, local community content, software, and/or any other content or data objects in electronic form.

(31) In some embodiments, metadata **124** comprises data about content **122**. For example, metadata **124** may include associated or ancillary information indicating or related to writer, director, producer, composer, artist, actor, summary, chapters, production, history, year, trailers, alternate versions, related content, applications, and/or any other information pertaining or relating to the content **122**. Metadata **124** may also or alternatively include links to any such information pertaining or relating to the content **122**. Metadata **124** may also or alternatively include one or more indexes of content **122**, such as but not limited to a trick mode index.

(32) The multimedia environment **102** may include one or more system servers **126**. The system servers **126** may operate to support the media devices **106** from the cloud. It is noted that the structural and functional aspects of the system servers **126** may wholly or partially exist in the same or different ones of the system servers **126**.

(33) The media devices **106** may exist in thousands or millions of media systems **104**. Accordingly, the media devices **106** may lend themselves to crowdsourcing embodiments and, thus, the system servers **126** may include one or more crowdsource servers **128**.

(34) For example, using information received from the media devices **106** in the thousands and millions of media systems **104**, the crowdsource server(s) **128** may identify similarities and overlaps between closed captioning requests issued by different users **132** watching a particular movie. Based on such information, the crowdsource server(s) **128** may determine that turning closed captioning on may enhance users' viewing experience at particular portions of the movie (for example, when the soundtrack of the movie is difficult to hear), and turning closed captioning off may enhance users' viewing experience at other portions of the movie (for example, when displaying closed captioning obstructs critical visual aspects of the movie). Accordingly, the crowdsource server(s) **128** may operate to cause closed captioning to be automatically turned on and/or off during future streaming's of the movie.

(35) The system servers **126** may also include an audio command processing module **130**. As noted above, the remote control **110** may include a microphone **112**. The microphone **112** may receive audio data from users **132** (as well as other sources, such as the display device **108**). In some embodiments, the media device **106** may be audio responsive, and the audio data may represent verbal commands from the user **132** to control the media device **106** as well as other components in the media system **104**, such as the display device **108**.

(36) In some embodiments, the audio data received by the microphone **112** in the remote control **110** is transferred to the media device **106**, which is then forwarded to the audio command processing module **130** in the system servers **126**. The audio command processing module **130** may operate to process and analyze the received audio data to recognize the user **132**'s verbal command. The audio command processing module **130** may then forward the verbal command back to the media device **106** for processing.

(37) In some embodiments, the audio data may be alternatively or additionally processed and analyzed by an audio command processing module **216** in the media device **106** (see FIG. 2). The media device **106** and the system servers **126** may then cooperate to pick one of the verbal commands to process (either the verbal command recognized by the audio command processing module **130** in the system servers **126**, or the verbal command recognized by the audio command processing module **216** in the media device **106**).

(38) FIG. 2 illustrates a block diagram of an example media device **106**, according to some embodiments. Media device **106** may include a streaming module **202**, processing module **204**,

storage/buffers **208**, and user interface module **206**. As described above, the user interface module **206** may include the audio command processing module **216**.

(39) The media device **106** may also include one or more audio decoders **212** and one or more video decoders **214**.

(40) Each audio decoder **212** may be configured to decode audio of one or more audio formats, such as but not limited to AAC, HE-AAC, AC3 (Dolby Digital), EAC3 (Dolby Digital Plus), WMA, WAV, PCM, MP3, OGG GSM, FLAC, AU, AIFF, and/or VOX, to name just some examples.

(41) Similarly, each video decoder **214** may be configured to decode video of one or more video formats, such as but not limited to MP4 (mp4, m4a, m4v, f4v, f4a, m4b, m4r, f4b, mov), 3GP (3gp, 3gp2, 3g2, 3gpp, 3gpp2), OGG (ogg, oga, ogv, ogx), WMV (wmv, wma, asf), WEBM, FLV, AVI, QuickTime, HDV, MXF (OPla, OP-Atom), MPEG-TS, MPEG-2 PS, MPEG-2 TS, WAV, Broadcast WAV, LXF, GXF, and/or VOB, to name just some examples. Each video decoder **214** may include one or more video codecs, such as but not limited to H.263, H.264, HEV, MPEG1, MPEG2, MPEG-TS, MPEG-4, Theora, 3GP, DV, DVCPro, DVCPro, DVCProHD, IMX, XDCAM HD, XDCAM HD422, and/or XDCAM EX, to name just some examples.

(42) Now referring to both FIGS. **1** and **2**, in some embodiments, the user **132** may interact with the media device **106** via, for example, the remote control **110**. For example, the user **132** may use the remote control **110** to interact with the user interface module **206** of the media device **106** to select content, such as a movie, TV show, music, book, application, game, etc. The streaming module **202** of the media device **106** may request the selected content from the content server(s) **120** over the network **118**. The content server(s) **120** may transmit the requested content to the streaming module **202**. The media device **106** may transmit the received content to the display device **108** for playback to the user **132**.

(43) In streaming embodiments, the streaming module **202** may transmit the content to the display device **108** in real time or near real time as it receives such content from the content server(s) **120**. In non-streaming embodiments, the media device **106** may store the content received from content server(s) **120** in storage/buffers **208** for later playback on display device **108**.

(44) Display Device Upgrades

(45) The technology as described herein, in some embodiments, may be configured to upgrade internal display device (e.g., television) hardware components for both streaming and non-streaming video and audio. In these embodiments, the technology may be applied broadly to any configurable aspect of a hardware replacement process. Display device **108** may be a monitor, television (TV), computer, touch screen, smart phone, tablet, wearable (such as a watch or glasses), virtual reality (VR) headset, appliance, internet of things (IOT) device, automotive display, gaming display, heads-up display (HUD), and/or projector, to name just a few examples.

(46) FIGS. **3A** and **3B** illustrate a block diagram of two configurations of display device **108** with original hardware **302**, according to some embodiments. Original hardware **302** may be implemented by processing logic that can comprise hardware (e.g., system on a chip (SOC), circuitry, dedicated logic, programmable logic, microcode, memory, etc.) and further include software (e.g., instructions executing on a processing device). In one non-limiting example embodiment, a processing device of original hardware **302** will receive instructions from an operating system (OS). An operating system is system software that manages the computer hardware, software resources, and provides common services for computer programs, such as streaming functionality. Example hardware is further described in association with FIG. **4**. It is to be appreciated that not all components may be needed to perform the disclosure provided herein. Further, some of the components described may include operational processes that may be performed simultaneously, or in a different order than described for FIGS. **3A** and **3B**, as will be understood by a person of ordinary skill in the art.

(47) The technology described may implement, for example, a television upgrade by connecting

new hardware **310** through existing display device **108** ports USB **304** or HDMI **306**. As with the original hardware **302**, in one non-limiting example embodiment, new hardware **310** is configured to receive instructions from an operating system (OS). This OS may be the same as an existing OS associated with the original hardware **302**, be an upgraded OS or be a new OS without departing from the scope of the technology disclosed herein. While the embodiments disclosed are directed to plug-in ports located on an external surface of display device **108**, any known or future port or interface may implement the technology described herein, including wireless interfaces, without departing from the scope of the technology described herein.

(48) The new hardware **310** may be configured in multiple embodiments (see FIGS. 5A-5D). In one embodiment, the new hardware replaces one or more hardware components of the display device original hardware. For example, the new hardware may be a new external tuner that replaces an original internal tuner. The terms “new hardware”, “external hardware”, “new hardware components”, “external computer device”, “external card”, “external stick” all refer to external hardware replacements, hardware upgrades or new functional hardware components as described throughout this disclosure.

(49) In some embodiments, the new hardware operates in conjunction with the display device original internal hardware. In this embodiment, the original and new hardware may take responsibility for specific tasks or share tasks. In one non-limiting example, the original hardware may retain display driver responsibility. A display driver is software that allows the operating system to work with the graphics hardware. In another non-limiting example, the original hardware may retain video and audio routing from source to display or to speakers. In another non-limiting example, the old and new hardware share WiFi responsibilities. In another non-limiting example, the new hardware can generate new display drivers.

(50) In some embodiments, the new hardware implements new functionality not previously available to the display device. For example, the new hardware includes a disco ball controller not previously available to the display device. In another example, the new hardware can process new data formats or new display or audio standards.

(51) Referencing FIG. 3A and FIG. 3B, Universal Serial Bus (USB) **304** is an industry standard that establishes specifications for cables, connectors and protocols for connection, communication and power supplies (interfacing) between computers, peripherals and other computers. A Consumer Electronics Control (CEC) capability allows USB devices to control each other when necessary and allows a user to operate multiple devices with one handheld remote control device. A broad variety of USB hardware exists, including many different connector types, of which USB-C is the most recent. However, the specific USB connector type does not limit the technology as described herein. Current and future port or interface standards may be substituted without departing from the scope of technology described herein.

(52) High-Definition Multimedia Interface (HDMI) **306** is a proprietary audio/video interface for transmitting uncompressed video data and compressed or uncompressed digital audio data from an HDMI-compliant source device, such as a display controller, to a compatible computer monitor, video projector, digital television, or digital audio device.

(53) HDMI implements the 861 standards, which define video formats and waveforms, transport of compressed and uncompressed audio, auxiliary data, and implementations of the Video Electronics Standards Association (VESA). A Consumer Electronics Control (CEC) capability allows HDMI devices to control each other when necessary and allows a user to operate multiple devices with one handheld remote control device. Several versions of HDMI have been developed and deployed since the initial release of the technology, but all use the same cable and connector. Other than improved audio and video capacity, performance, resolution and color spaces, newer versions have optional advanced features such as 3D, Ethernet data connection, and CEC extensions. However, specific HDMI connector types and standards do not limit the technology as described herein. Current and future connectors and standards may be substituted without departing from the scope

of technology described herein.

(54) In particular, the figures illustrate a display device **108** (e.g., television) with a card bay, shown as HDMI and USB ports, to receive a plug-in card or stick, such as new hardware card **310**, the configuration of which is discussed further below. Display device **108** is operative with remote **110**, which includes buttons and or voice input control. Alternatively, or in addition to, remote **110** may include home voice interface devices connected to the internet. The display device **108** is an example of a device that is capable of presenting audio and visual information. Such a device may also be commonly referred to as an audio/visual (A/V) device. Original hardware component **302** may be configured to interact with remote control **110** through commands. The remote control **110** can be any component, part, apparatus and/or method for controlling the media device **106** and/or display device **108**. In an embodiment, the remote control **110** wirelessly communicates with the media device **106** or display device **108** using cellular, Bluetooth, infrared, etc., or any combination thereof.

(55) Media device **106** may be a streaming media device, Digital Video Disc (DVD) or BLU-RAY device, audio/video playback device, cable box, and/or digital video recording device, to name just a few examples. For illustrative purposes only, media device **106** is shown communicating commands and content to/from the original hardware, but not connected. In some embodiments, media device **106** can be a part of, integrated with, operatively coupled to, and/or connected to its respective display device **108**. In practice, the media device **106** may be connected to the display device **108** through the USB port **304**, HDMI port **306**, coaxial cable connections, wirelessly connected or other known or future connection arrangements. Connecting the new hardware to the display device automatic enables of the Universal Serial Bus (USB) CEC or HDMI CEC based on the connection of the media device to the display device through a respective USB or HDMI port.

(56) Each media device **106** may be configured to communicate with a network via a communication device. The communication device may include, for example, a cable modem or satellite TV transceiver. The media device **106** may communicate with the communication device over a link, wherein the link may include wireless (such as WiFi) and/or wired connections.

(57) An interface module **308** is also connected to the original hardware **302**. In one non-limiting embodiment, the interface module **308** is communicatively coupled to a processor **402** (see FIG. 4). The interface module is a set of executable instructions stored in memory, which are executed by the processor **402**. The interface module controls communications between the original hardware and the new hardware connected through the HDMI, USB or wireless ports or interfaces.

(58) FIG. 3A shows the new hardware **310** in an unconnected state. However, FIG. 3B, illustrates when the new hardware **310** is operative with the display device **108**. In some embodiments, the new hardware may operate as a master device, while the display device operates as a slave device (or peripheral device). Put another way, the new hardware **310** performs one or more functions originally performed by one or more components of the original hardware **302**. Commands intended for the original hardware **302** are rerouted to the new hardware for processing. In one non-limiting example, control signals (e.g., commands) to the display device **108** may be supplied through interface (I/O) **308** to the new hardware **310** with CEC commands in the HDMI protocol (when new hardware **310** is connected to the HDMI port as shown in FIG. 3B). For instance, the new hardware **310** communicates with the original hardware (e.g., with an internal controller) with a call or pull operation over, for example, an internal HDMI CEC link. Alternatively, or in addition to, the display device forwards the command as a push over an internal HDMI CEC link to the new hardware **310**. The new hardware **310** may then provide the appropriate command back to the display device **108** via the HDMI CEC link. While described for commands, display and audio signals may be generated or controlled by the new hardware in a similar fashion. In one non-limiting example, a media device **106** connected to the display device may be controlled by the new hardware **310**. Alternatively, or in addition to, the original hardware receives media content for play on display device **108** and forwards the media content to internal display and audio device

drivers for subsequent display with associated audio based, at least in part, on commands from the new hardware. For example, using the earlier example, remote control commands to change the channel would be forwarded to the new external tuner and resulting commands from the new external tuner would be communicated to the display device. While described for HDMI, the technology described herein may be applied to USB ports **304**, wirelessly or with any known or future coupling technology without departing from the scope of the technology described herein.

(59) The new hardware **310** may, in some embodiments be configured with media streaming capabilities. For example, the new hardware **310** may provide an internet streaming solution as an add-on to a television. Consequently, the television manufacturer does not need to bear the cost of an internet streaming solution associated with a particular supplier. Alternatively, or in addition to, a consumer may choose from one of many internet access cards and simply plug a selected solution into the display device **108**. The internet access card would, in some embodiments, communicate with the new hardware **310** to control access and display of the streaming video and audio signals.

(60) Thus, the new hardware card **310** provides an external hardware upgrade device solution as an add-on to, for example, a television. When a television is purchased by a customer, methods for improving the experience or performance of the TV are generally limited to software only updates. However, the various embodiments described herein provide a technical improvement to the television itself. In addition, the new hardware may, in some embodiments, add new functionality to the television not envisioned by the Original Equipment Manufacturer (OEM) at the time of design or build. The technology described herein overcomes these limitations and therefore solves a problem that exists in the production cycle of existing display devices.

(61) FIG. **4** is a block diagram of original hardware **302** for display device **108**, according to some embodiments. Original hardware **302** may be implemented by processing logic (e.g., processor **402**) that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device) stored in storage/buffers **404**, or a combination thereof. It is to be appreciated that not all components may be needed to perform the disclosure provided herein as will be understood by a person of ordinary skill in the art.

(62) The display device **108** is configured to include, for example, an external card bay electrical interface with physical card or cable connector receiving sections or ports (e.g., HDMI **420**, USB **422**, Audio/Video components, optical connectors, or coaxial connectors). In addition, the external card bay electrical interface may communicate with internal device interface **416** that may include circuitry and software components. For example, device interface **416** (e.g., television interfaces) may include various encoding and decoding components to communicate video, audio, control and network signals as is known to one skilled in the art and therefore will not be described in greater detail herein.

(63) In addition, original hardware **302** may communicate wirelessly **424** with one or more external devices. The processor **402** may be connected to a network interface **412**, such as, a WiFi circuit, which may include one or more antennas. The WiFi circuit supports wireless communications **424**, with for example, a wireless router. An Ethernet port, not shown, may also be provided, which allows for a hardwired connection to the internet. In various embodiments, one or more of the ports, connectors and interfaces should be exposed to the hardware upgrade **310** when it is inserted or otherwise coupled to the display device **108**.

(64) Other inputs for display device **108** may include a tuner **410** (e.g., a high definition television tuner) providing tuning of video and audio signals. A user interface **418** communicates with a remote control **426** (same as **110**) that can be any component, part, apparatus and/or method for controlling the media device **106** and/or display device **108**, such as a remote control, a tablet, laptop computer, smartphone, wearable, on-screen controls, integrated control buttons, audio controls, or any combination thereof, to name just a few examples. In an embodiment, the remote control **426** wirelessly communicates, through user interface **418**, with the media device **106** and/or display device **108** using cellular, Bluetooth, infrared, etc., or any combination thereof. The remote

control **426** may include a microphone, as previously described. A display controller **414** operates under the control of the processor **402** to select an audio/visual signal input. The processor **402** performs standard functions to produce a video output **406** and an audio output **408** (e.g., through one or more at speakers). The display device **108** also includes a standard power management circuit **409**.

(65) The card bay provides a link to the internal electronics of the display device **108**. This allows the hardware upgrade **310** to control the display device **108** as further described in various embodiments. The hardware upgrade may be to any or all of the components described in the original hardware. In addition, the one or more components included in the hardware upgrade **310** may be same, upgraded or new, previously unavailable, hardware components. While described for hardware upgrades, the new hardware upgrade **310** may also include same, upgraded or a new operating system (OS). As previously described, hardware upgrade may also implement video and audio streaming technology.

(66) FIG. 5A-5D collectively illustrate a plurality of configurations for display device software layers (i.e., software stack) for internal and external hardware. Operations described may be implemented by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. It is to be appreciated that not all components may be needed to perform the disclosure provided herein. Further, some of the components may be performed simultaneously, or in a different order than described for FIG. 5A-5D, as will be understood by a person of ordinary skill in the art.

(67) FIG. 5A illustrates a block diagram of one non-limiting example embodiment where a display device **108** has yet to initiate operational functionality with an external hardware upgrade. In this example embodiment, the software layers are functional for existing original hardware **302**. For example, the external hardware may not be connected or may be connected, but may not have initiated or completed a process to take existing control or data functionality for one or more original hardware components. In this configuration, an original operating system (OS) **508** provides instructional processing and control of display and audio functionality of the display device **108**. The original OS will be stored in storage **404** and be processed by processor **402**. For example, a TV operating system makes it possible for a user to access and control advanced features and connected devices. The TV operating system allows a user to browse not just channels on satellite or cable TV, but also on demand video services. The system also provides access pictures, music or video content on connected storage devices or streams. The TV operating systems may be open or closed source.

(68) System Management **510** provides configuration, monitoring and performance management of the display device. The system management provides management across the display device **108** delivery infrastructure. For example, the system management may aggregate data from various sources such as devices, network applications and integrated third party applications.

(69) Rendering component **512** generates images or motion pictures from video data. In one embodiment, the video data is streamed to the display device, stored in storage/buffers **404**, and provided to the display screen through display drivers. Rendering component **512** may be driven by processor **402** or be processed by a stand-alone graphics engine (not shown). Graphics pipelines, found in most graphics processing units (GPUs), which consist of multiple arithmetic units, or complete CPUs, implement the various stages of common rendering operations (perspective projection, window clipping, color and light calculation, rendering, etc.).

(70) Media pipeline **514** may organize and order a sequence of video and audio to be rendered on the display screen and output from built-in or external sound sources (e.g., speakers). Buffering may be needed between the media receiving and playing stages as download speeds and quality may vary, or when media items may be created, lost or destroyed along the media pipeline. Buffering may also be needed to accommodate irregularities in the rates at which the application

feeds items to the rendering stage and the rate at which the media is consumed (displayed).

(71) Hardware abstraction layers **516** are sets of routines in software that provide programs with access to hardware resources through programming interfaces. The programming interface allows devices in a particular class of hardware devices to be accessed through identical interfaces even though they may contain different subclasses of devices that each provide a different hardware interface. The hardware abstraction layer resides above the application programming interface (API) in a software stack, whereas the application layer (often written in a high level language) resides above the API and communicates with the hardware by calling functions in the API.

(72) Hardware abstraction layers **516** often allow programmers to write device-independent, high performance applications by providing standard operating system (OS) calls to hardware. The process of abstracting pieces of hardware is often done from the perspective of a CPU. Each type of CPU has a specific Instruction Set Architecture (ISA). The ISA represents the primitive operations of the machine that are available for use by assembly programmers and compiler writers. The same type of abstraction may be made in operating systems, but OS APIs now represent the primitive operations of the machine, rather than an ISA. This allows a programmer to use OS-level operations (e.g. task creation/deletion) in their programs while retaining portability over a variety of different platforms.

(73) Original hardware **302** has been previously described in association with FIG. **4**. The original hardware **302** may include the components as described in FIG. **4** or may include more or less components (e.g., tuner, interfaces, processor(s), internal controller(s), storage, buffers, and video or audio I/O, et al.) based on the display device design. An API exposes the original hardware **302** to the upper layers to realize communication of data and commands.

(74) Media device **106** may be configured as peripheral equipment to the display device. **108**. Media device **106** may be a streaming media device, DVD or BLU-RAY device, audio/video playback device, cable box, and/or digital video recording device, to name just a few examples. Display device **108** may be a monitor, television (TV), computer, touch screen, smart phone, tablet, wearable (such as a watch or glasses), virtual reality (VR) headset, appliance, internet of things (IOT) device, automotive display, gaming display, heads-up display (HUD), and/or projector, to name just a few examples. In some embodiments, media device **106** can be a part of, integrated with, operatively coupled to, and/or connected to its respective display device **108**. As media device **106** acts as a peripheral to the display device **108**, a PAL API exposes the media device **106** to the upper layers to realize communication of data and commands.

(75) As shown, external hardware **518** is not yet operational with the display device original hardware. For example, the external hardware has yet to expose one or more replacement or upgrade hardware components to the hardware abstraction layer through an API. External hardware **518** may be configured with one or more replacement, upgraded components or new components not part of the original hardware as previously shown and described in association with FIG. **4**. For example, a new tuner may represent an upgrade to an original tuner based on any of, but not limited to, performance, features, encoding/decoding mechanisms, drivers, interfaces, etc.

(76) FIG. **5B** illustrates a block diagram of one non-limiting example embodiment where a display device **108** has one or more original hardware **302** components replaced by one or more external hardware components **518** (same as **310**). In this exemplary embodiment, the external hardware includes one or more upgraded components to directly replace the original hardware component. Alternatively, or in addition to, in a repair scenario, the external hardware component may replace the existing component with an operational replacement, but not be an upgrade.

(77) In this example embodiment, the external hardware **518** is connected by a ported connection or wireless interface. The connection establishes communication between the internal and external hardware devices. In this connected configuration, the software layers are functional for both the existing original hardware **302** as well as any replacement components that are part of external hardware **518**. External hardware upgrade **518** initiates operational functions with hardware and

software of display device **108** by importing a hardware configuration and one or more of the internal hardware settings of the original hardware **302**. For example, the external hardware may import all original hardware settings or only import settings for an external hardware upgrade component. The original hardware settings may include, but not be limited to, picture settings, audio settings, network configurations, display drivers, buffer size, download bitrates, display hardware refresh rates, audio and video sources (e.g., peripherals or data streaming sources), interface configurations, etc.

(78) In this configuration, an original operating system (OS) **508** may provide instructional control of display and audio functionality of the display device **108**. The original OS will be stored in storage **404** and be processed by processor **402**. Alternatively, or in addition to, a new OS **522** stored on computer memory within external hardware **518**, may provide instructional control of one or more tasks of the original hardware, the external hardware **518** or both. For example, the new OS may acquire task responsibility for WiFi processing.

(79) Original hardware **302** has been previously described in association with FIG. 4. The original hardware **302** may include the components as described in FIG. 4 or may include more or less components (e.g., tuner, interfaces, processor(s), internal controller(s), storage, buffers, and video or audio I/O, et al.) based on the display device design. However, in this configuration, the original hardware acts as a peripheral to the external hardware **518**. As such, the API exposing the original hardware **302** to the upper layers to realize communication of data and commands is modified with a PAL API. Commands received by the original hardware are forwarded or rerouted to the external hardware new API. More specifically, commands intended for an original internal hardware component are rerouted to the replacement hardware component of external hardware **518**.

Rerouting is implemented as calls from the external hardware **518** or as pushes from the original hardware. For example, hardware abstraction layers **516** provide operating system (OS) calls to the new replacement hardware component. Calls may include, but are not limited to remote control inputs (voice or key clicks), hardware events, audio return channels, Cable TV (CATV) information, tuner requests, data streaming requests, etc. The commands may include hardware abstraction messages from the internal controller.

(80) In one non-limiting example, the external hardware may recognize, based on an internal display device platform configuration, an internal controller within the internal display device platform, wherein the internal controller is configured to render media content on the display device based on display device commands. The internal controller may be implemented with a host controller interface (HCI) and receiving control of the internal controller is based on enabling a USB host controller for the HCI.

(81) The external hardware receives, from the internal controller, display device commands bound for the internal display device platform. The hardware configuration settings may include, but not be limited to, picture settings, audio settings, network configurations, display drivers, buffer size, download bitrates, display hardware refresh rates, audio and video sources (e.g., peripherals or data streaming sources), interface configurations, etc. The display device commands may be rerouted to the external hardware based on a push instruction from the display device or based on a call instruction from the external hardware.

(82) Media device **106** may be configured as peripheral equipment to the display device. **108**. Media device **106** may be a streaming media device, DVD or BLU-RAY device, audio/video playback device, cable box, and/or digital video recording device, to name just a few examples. Display device **108** may be a monitor, television (TV), computer, touch screen, smart phone, tablet, wearable (such as a watch or glasses), virtual reality (VR) headset, appliance, internet of things (IoT) device, automotive display, gaming display, heads-up display (HUD), and/or projector, to name just a few examples. In some embodiments, media device **106** can be a part of, integrated with, operatively coupled to, and/or connected to its respective display device **108**. As media device **106** acts as a peripheral to the display device **108**, a PAL API exposes the media device **106**

to the upper layers to realize communication of data and commands. External hardware **518** may communicate and control the media device as a peripheral using, for example, a Consumer Electronics Control (CEC) capability over a bi-directional communications channel through interface **308**.

(83) As shown, external hardware **518** is operational with the display device original hardware. For example, the external hardware exposes one or more replacement or upgrade hardware components to the hardware abstraction layer **516** through a new API. The one or more replacement or upgrade hardware components take over task responsibilities for the original hardware component without a user having to provide any additional inputs other than plugin the new stick, card or connect to the new hardware wirelessly.

(84) FIG. 5C illustrates a block diagram of one non-limiting example embodiment where a display device **108** original hardware **302** shares operational functionality with an external hardware upgrade. In this example embodiment, the external hardware includes one or more upgraded components to share responsibilities with an original hardware component. In a non-limiting example, the external hardware component is an upgraded WiFi component and the original and external component share processing responsibilities for the display device.

(85) In this example embodiment, the external hardware **518** is connected by a ported connection or wireless interface. The connection establishes communication between the internal and external hardware devices. In this connected configuration, the software layers are functional for both the existing original hardware **302** as well as any replacement components that are part of external hardware **518**. External hardware upgrade **518** initiates operational functions with hardware and software of display device **108** by importing one or more of the internal hardware settings of the original hardware **302**. For example, the external hardware may import all original hardware settings or only import settings for an external hardware upgrade component. The original hardware settings may include, but not be limited to, picture settings, audio settings, network configurations, display drivers, buffer size, download bitrates, display hardware refresh rates, audio and video sources (e.g., peripherals or data streaming sources), interface configurations, etc.

(86) In this configuration, an original operating system (OS) **508** may provide instructional control of display and audio functionality of the display device **108**. The original OS will be stored in storage **404** and be processed by processor **402**. Alternatively, or in addition to, a new OS **522** stored on computer memory within external hardware **518**, may provide instructional control of one or more tasks of the original hardware, the external hardware **518** or both. For example, the new OS may acquire task responsibility for WiFi processing.

(87) Original hardware **302** has been previously described in association with FIG. 4. The original hardware **302** may include the components as described in FIG. 4 or may include more or less components (e.g., tuner, interfaces, processor(s), internal controller(s), storage, buffers, and video or audio I/O, et al.) based on the display device design. However, in this configuration, the original and new external hardware act as peripherals to the display device **108**. As such, the API exposing the original hardware **302** to the upper layers to realize communication of data and commands is modified with a PAL API. Commands received by the original hardware are forwarded or rerouted to the external hardware new API. More specifically, commands intended for an original internal hardware component are rerouted to the shared hardware component of external hardware **518**. Rerouting is implemented as calls from the external hardware **518** or as pushes from the original hardware. For example, hardware abstraction layers **516** provide operating system (OS) calls to the new replacement hardware component. Calls may include, but are not limited to remote control inputs (voice or key clicks), hardware events, audio return channels, CATV information, tuner requests, data streaming requests, etc.

(88) Media device **106** may be configured as peripheral equipment to the display device. **108**. Media device **106** may be a streaming media device, DVD or BLU-RAY device, audio/video playback device, cable box, and/or digital video recording device, to name just a few examples.

Display device **108** may be a monitor, television (TV), computer, touch screen, smart phone, tablet, wearable (such as a watch or glasses), virtual reality (VR) headset, appliance, internet of things (IOT) device, automotive display, gaming display, heads-up display (HUD), and/or projector, to name just a few examples. In some embodiments, media device **106** can be a part of, integrated with, operatively coupled to, and/or connected to its respective display device **108**. As media device **106** acts as a peripheral to the display device **108**, a PAL API exposes the media device **106** to the upper layers to realize communication of data and commands. External hardware **518** may communicate and control the media device as a peripheral using, for example, a Consumer Electronics Control (CEC) capability over a bi-directional communications channel through interface **308**.

(89) As shown, external hardware **518** is operational with the display device original hardware. For example, the external hardware exposes one or more shared hardware components to the hardware abstraction layer **516** as a peripheral through a PAL API. The one or more shared hardware components share one or more task responsibilities for the original hardware component without a user having to provide any additional inputs other than plugin the new stick, card or connect to the new hardware wirelessly.

(90) FIG. 5D illustrates a block diagram of one non-limiting example embodiment where a display device **108** receives new operational functionality with new external hardware **520** that includes one or more new hardware components not previously part of the original hardware. For example, a television with a 4K tuner may not be able to display a new data format (e.g., 128K) when it becomes available. In this embodiment, the external hardware includes one or more new components (e.g., new 128K tuner) to provide new functionality to display device **108**.

(91) In this example embodiment, the external hardware **518** is connected by a ported connection or wireless interface. The connection establishes communication between the internal and external hardware devices. In this connected configuration, the software layers are functional for both the existing original hardware **302** as well as any new hardware components that are part of external hardware **518**. External hardware **518** initiates operational functions with hardware and software of display device **108** by importing one or more of the internal hardware settings of the original hardware **302**. For example, the external hardware may import all original hardware settings or only import settings for an external hardware component. The original hardware settings may include, but not be limited to, picture settings, audio settings, network configurations, display drivers, buffer size, download bitrates, display hardware refresh rates, audio and video sources (e.g., peripherals or data streaming sources), interface configurations, etc.

(92) In this configuration, an original operating system (OS) **508** may provide instructional control of display and audio functionality of the display device **108**. The original OS will be stored in storage **404** and be processed by processor **402**. In addition, a new OS **522** stored on computer memory within external hardware **518**, may provide instructional control of one or more tasks of the original hardware, the external hardware **518** new hardware component or both. For example, the new OS may acquire task responsibility for display drivers for a new display format or standard.

(93) Original hardware **302** has been previously described in association with FIG. 4. The original hardware **302** may include the components as described in FIG. 4 or may include more or less components (e.g., tuner, interfaces, processor(s), internal controller(s), storage, buffers, and video or audio I/O, et al.) based on the display device design. However, in this configuration, the original hardware acts as a peripheral to the external hardware **518**. As such, the API exposing the original hardware **302** to the upper layers to realize communication of data and commands is modified with a PAL API. Commands received by the original hardware are forwarded or rerouted to the external hardware new API. More specifically, commands intended for an original internal hardware component are rerouted to the new hardware component of external hardware **518**. Rerouting is implemented as calls from the external hardware **518** or as pushes from the original hardware. For

example, hardware abstraction layers **516** provide operating system (OS) calls to the new replacement hardware component. Calls may include, but are not limited to remote control inputs (voice or key clicks), hardware events, audio return channels, CATV information, tuner requests, data streaming requests, etc.

(94) Media device **106** may be configured as peripheral equipment to the display device. **108**. Media device **106** may be a streaming media device, DVD or BLU-RAY device, audio/video playback device, cable box, and/or digital video recording device, to name just a few examples. Display device **108** may be a monitor, television (TV), computer, touch screen, smart phone, tablet, wearable (such as a watch or glasses), virtual reality (VR) headset, appliance, internet of things (IOT) device, automotive display, gaming display, heads-up display (HUD), and/or projector, to name just a few examples. In some embodiments, media device **106** can be a part of, integrated with, operatively coupled to, and/or connected to its respective display device **108**. As media device **106** acts as a peripheral to the display device **108**, a PAL API exposes the media device **106** to the upper layers to realize communication of data and commands. External hardware **518** may communicate and control the media device as a peripheral using, for example, a Consumer Electronics Control (CEC) capability over a bi-directional communications channel through interface **308**.

(95) As shown, external hardware **518** is operational with the display device original hardware. For example, the external hardware exposes one or more new hardware components to the hardware abstraction layer **516** through a new API, as modified for the new component. The one or more new hardware components implement their associated task responsibilities without a user having to provide any additional inputs other than plugin the new stick, card or connect to the new hardware wirelessly. External hardware **518** may be configured with one or more new components not part of the original hardware as previously shown and described in association with FIG. **4**. However, one or more components of the original hardware **302** may be upgraded, replaced by the newly available hardware component features.

(96) FIG. **6** illustrates a flow diagram of the process **600** for replacing, sharing or adding new hardware by adding external hardware for a display device. While described for a display device, the new hardware may be operational for any media device upgrade.

(97) In response to connection of the external hardware to a display device, in **602**, the external hardware establishes a communication channel between an interface of the external hardware and an interface of the original internal hardware. For example, in an HDMI connection, external hardware may communicate and control the display device as a peripheral using, for example, a Consumer Electronics Control (CEC) capability over a bi-directional communications channel through an interface.

(98) In **604**, after a communication channel has been established (e.g., HDMI CEC), the external hardware automatically requests, from the display device, a hardware configuration of the display device. For example, the external hardware may recognize, based on an internal display device platform configuration, an internal controller within the internal display device platform, wherein the internal controller is configured to render media content on the display device based on display device commands. The hardware configuration settings may include, but not be limited to, picture settings, audio settings, network configurations, display drivers, buffer size, download bitrates, display hardware refresh rates, audio and video sources (e.g., peripherals or data streaming sources), interface configurations, etc.

(99) In **606**, the external hardware may request control of the internal controller to redirect internal commands intended for the original hardware to the external hardware. The redirect may be implemented as a push or pull (e.g., call) operation.

(100) In **608**, the external hardware receives, from the internal controller, one or more display device commands bound for the internal display device platform. The display device commands may be call operations that include HAL instructions. For example, the external hardware is an

upgraded tuner and the external hardware receives remote control commands requests to tune to a new channel.

(101) In **610**, the external hardware communicates commands to the display device for operation of one or more hardware components (e.g., internal hardware components, external hardware components or both) for rendering the media data (and outputting associated audio) on the display device.

(102) In **612**, the external hardware manages, at least in part, the display device using the commands from the external hardware. For example, using the earlier example, the external tuner directs a media source to display the newly tuned channel video based on an original remote control command to change to that channel. All described video actions may be equally applicable to audio.

(103) While described herein for hardware replacements or upgrades, one or more software functions of an existing display device may be upgraded as well without departing from the scope of the technology described herein.

(104) Example Computer System

(105) Various embodiments may be implemented, for example, using one or more well-known computer systems, such as computer system **700** shown in FIG. 7. Also or alternatively, one or more computer systems **700** may be used, for example, to implement any of the embodiments discussed herein, as well as combinations and sub-combinations thereof.

(106) Computer system **700** may include one or more processors (also called central processing units, or CPUs), such as a processor **704**. Processor **704** may be connected to a communication infrastructure or bus **706**.

(107) Computer system **700** may also include user input/output device(s) **703**, such as monitors, keyboards, pointing devices, etc., which may communicate with communication infrastructure **706** through user input/output interface(s) **702**.

(108) One or more of processors **704** may be a graphics processing unit (GPU). In an embodiment, a GPU may be a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU may have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.

(109) Computer system **700** may also include a main or primary memory **708**, such as random access memory (RAM). Main memory **708** may include one or more levels of cache. Main memory **708** may have stored therein control logic (i.e., computer software) and/or data.

(110) Computer system **700** may also include one or more secondary storage devices or memory **710**. Secondary memory **710** may include, for example, a hard disk drive **712** and/or a removable storage device or drive **714**. Removable storage drive **714** may be a floppy disk drive, a magnetic tape drive, a compact disk drive, an optical storage device, tape backup device, and/or any other storage device/drive.

(111) Removable storage drive **714** may interact with a removable storage unit **718**. Removable storage unit **718** may include a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **718** may be a floppy disk, magnetic tape, compact disk, DVD, optical storage disk, and/or any other computer data storage device. Removable storage drive **714** may read from and/or write to removable storage unit **718**.

(112) Secondary memory **710** may include other means, devices, components, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **700**. Such means, devices, components, instrumentalities or other approaches may include, for example, a removable storage unit **722** and an interface **720**. Examples of the removable storage unit **722** and the interface **720** may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB or other port, a memory

card and associated memory card slot, and/or any other removable storage unit and associated interface.

(113) Computer system **700** may further include a communication or network interface **724**. Communication interface **724** may enable computer system **700** to communicate and interact with any combination of external devices, external networks, external entities, etc. (individually and collectively referenced by reference number **728**). For example, communication interface **724** may allow computer system **700** to communicate with external or remote devices **728** over communications path **726**, which may be wired and/or wireless (or a combination thereof), and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from computer system **700** via communication path **726**.

(114) Computer system **700** may also be any of a personal digital assistant (PDA), desktop workstation, laptop or notebook computer, netbook, tablet, smart phone, smart watch or other wearable, appliance, part of the Internet-of-Things, and/or embedded system, to name a few non-limiting examples, or any combination thereof.

(115) Computer system **700** may be a client or server, accessing or hosting any applications and/or data through any delivery paradigm, including but not limited to remote or distributed cloud computing solutions; local or on-premises software (“on-premise” cloud-based solutions); “as a service” models (e.g., content as a service (CaaS), digital content as a service (DCaaS), software as a service (SaaS), managed software as a service (MSaaS), platform as a service (PaaS), desktop as a service (DaaS), framework as a service (FaaS), backend as a service (BaaS), mobile backend as a service (MBaaS), infrastructure as a service (IaaS), etc.); and/or a hybrid model including any combination of the foregoing examples or other services or delivery paradigms.

(116) Any applicable data structures, file formats, and schemas in computer system **700** may be derived from standards including but not limited to JavaScript Object Notation (JSON), Extensible Markup Language (XML), Yet Another Markup Language (YAML), Extensible Hypertext Markup Language (XHTML), Wireless Markup Language (WML), MessagePack, XML User Interface Language (XUL), or any other functionally similar representations alone or in combination. Alternatively, proprietary data structures, formats or schemas may be used, either exclusively or in combination with known or open standards.

(117) In some embodiments, a tangible, non-transitory apparatus or article of manufacture comprising a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon may also be referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **700**, main memory **708**, secondary memory **710**, and removable storage units **718** and **722**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **700** or processor(s) **704**), may cause such data processing devices to operate as described herein.

(118) Based on the teachings contained in this disclosure, it will be apparent to persons skilled in the relevant art(s) how to make and use embodiments of this disclosure using data processing devices, computer systems and/or computer architectures other than that shown in FIG. 7. In particular, embodiments can operate with software, hardware, and/or operating system implementations other than those described herein.

CONCLUSION

(119) It is to be appreciated that the Detailed Description section, and not any other section, is intended to be used to interpret the claims. Other sections can set forth one or more but not all exemplary embodiments as contemplated by the inventor(s), and thus, are not intended to limit this disclosure or the appended claims in any way.

(120) While this disclosure describes exemplary embodiments for exemplary fields and applications, it should be understood that the disclosure is not limited thereto. Other embodiments and modifications thereto are possible, and are within the scope and spirit of this disclosure. For

example, and without limiting the generality of this paragraph, embodiments are not limited to the software, hardware, firmware, and/or entities illustrated in the figures and/or described herein. Further, embodiments (whether or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein.

(121) Embodiments have been described herein with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. Also, alternative embodiments can perform functional blocks, steps, operations, methods, etc. using orderings different than those described herein.

(122) References herein to “one embodiment,” “an embodiment,” “an example embodiment,” or similar phrases, indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such feature, structure, or characteristic into other embodiments whether or not explicitly mentioned or described herein. Additionally, some embodiments can be described using the expression “coupled” and “connected” along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, some embodiments can be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, can also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

(123) The breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A computing device, comprising: a memory; and one or more processing devices configured to execute instructions from the memory including instructions to: detect a connection of the computing device to an external High-Definition Multimedia Interface (HDMI) port of a display device; upon detecting the connection, request from the display device a hardware configuration of internal hardware of the display device; recognize, based on the hardware configuration, an internal controller within the display device, wherein the internal controller is configured to render media content on the display device based on first display device commands; request, based on a communication from the computing device to the display device, control of the internal controller; upon receiving control of the internal controller, generate a hardware call to redirect the first display device commands originating from the internal controller to the computing device; and communicate, to the internal controller and based on processing one or more of the redirected first display device commands, second display device commands based, at least in part, on the computing device replacing one or more hardware components of the internal hardware of the display device.
2. The computing device of claim 1, wherein the first display device commands include hardware abstraction messages forwarded from the internal controller.
3. The computing device of claim 1, further comprising one or more operating system upgrades to display device resident software operative with the internal hardware of the display device.
4. The computing device of claim 1, wherein the hardware configuration of the internal hardware

of the display device further comprises settings of the display device.

5. The computing device of claim 1, wherein the display device comprises a television.

6. The computing device of claim 1 wherein the external HDMI port is configured to communicate the first display device commands and the second display device commands based on an HDMI Consumer Electronics Control (CEC) specification.

7. The computing device of claim 6, wherein the one or more processing devices are configured to, upon detecting the connection, automatically enable communications based on the HDMI CEC specification.

8. The computing device of claim 1, wherein the computing device further comprises one or more display hardware components that are absent from the internal hardware of the display device.

9. The computing device of claim 1, wherein the internal controller comprises a host controller interface (HCI) and the receiving control of the internal controller is based on enabling an HDMI host controller for the HCI.

10. The computing device of claim 1, further comprising a wireless multimedia interface configured to wirelessly communicate the first display device commands and the second display device commands based on a wireless communication specification.

11. A computer-implemented method for upgrading a display device, the computer-implemented method comprising: detecting, by at least one computer processor, a connection of a computing device to an external High-Definition Multimedia Interface (HDMI) port of the display device; upon detecting the connection, requesting from the display device a hardware configuration of internal hardware of the display device; recognizing, based on the hardware configuration, an internal controller within the display device, wherein the internal controller is configured to render media content on the display device based on first display device commands; requesting, based on a communication from the computing device to the display device, control of the internal controller; upon receiving control of the internal controller, generating a hardware call to redirect the first display device commands originating from the internal controller to the computing device; and communicating, to the internal controller and based on processing one or more of the redirected first display device commands, second display device commands based, at least in part, on the computing device replacing one or more hardware components of the internal hardware of the display device.

12. The computer-implemented method of claim 11, wherein the external HDMI port is configured to communicate the first display device commands and the second display device commands based on an HDMI Consumer Electronics Control (CEC) specification.

13. The computer-implemented method of claim 11, wherein the first display device commands include hardware abstraction messages forwarded from the internal controller.

14. The computer-implemented method of claim 11, wherein the processing further comprises processing the one or more redirected first display device commands with one or more operating system upgrades for the display device.

15. The computer-implemented method of claim 11, wherein the hardware configuration of the internal hardware of the display device further comprises settings of the display device.

16. The computer-implemented method of claim 11, wherein the display device comprises a television.

17. A non-transitory computer-readable medium having instructions stored thereon that, when executed by at least one computing device, cause the at least one computing device to perform operations comprising: detecting a connection of the at least one computing device to an external High-Definition Multimedia Interface (HDMI) port of a display device; upon detecting the connection, requesting from the display device a hardware configuration of internal hardware of the display device; recognizing, based on the hardware configuration, an internal controller within the display device, wherein the internal controller is configured to render media content on the display device based on first display device commands; requesting, based on a communication from the at

least one computing device to the display device, control of the internal controller; upon receiving control of the internal controller, generating a hardware call to redirect the first display device commands originating from the internal controller to the at least one computing device; and communicating, to the internal controller and based on processing one or more of the redirected first display device commands, second display device commands based, at least in part, on the at least one computing device replacing one or more hardware components of the internal hardware of the display device.

18. The non-transitory computer-readable medium of claim 17, wherein the connection of the external HDMI port further comprises enabling communications based on an HDMI Consumer Electronics Control (CEC) specification.

19. The non-transitory computer-readable medium of claim 17, wherein the first display device commands include hardware abstraction messages forwarded from the internal controller.

20. The non-transitory computer-readable medium of claim 17, the operations further comprising: processing the one or more redirected first display device commands with one or more operating system upgrades for the display device.
